

BFGI Alumni Portal — Backend Server & Cost Analysis

Architectural & Financial Comparison: On-Campus Dedicated College PC vs. External Cloud VPS

Target Users 4,000 – 5,000 Registered	Peak Traffic 50 – 200 Concurrent	Date July 2026	Version 2.1
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Minimum Infrastructure Requirements

Frontend Hosting	Minimum vCPU / RAM	Storage Requirement	Monthly Bandwidth
Vercel (Free Tier)	2 vCPUs / 4 GB RAM	40 GB – 60 GB NVMe SSD	1 TB – 2 TB Transfer

Hosting Options

Option 1: On-Campus Dedicated College PC

RECOMMENDED

The college provides one dedicated desktop CPU/PC on campus (in a lab or server room), connected to 24×7 power (with UPS backup) and internet. By utilizing existing campus IT infrastructure, server hardware, hosting, electricity, and bandwidth are provided at no additional cost. Only a domain name needs to be purchased separately.

Server Hosting Cost: **Rs. 0 (Free)** • Domain Cost: **Rs. 499 - Rs. 999 / year** • Total Yearly Cost: **Rs. 499 - Rs. 999 / year**

Option 2: External Cloud VPS

ALTERNATIVE

Hosting on an external Cloud Virtual Private Server (VPS) provides a 99.9% cloud SLA uptime guarantee without requiring on-campus physical maintenance or local power/internet management. However, it requires an ongoing annual server hosting fee in addition to domain registration.

Cloud Server Providers (2 vCPU, 4–8 GB RAM)

Cloud Provider & Plan	Configuration	Monthly Cost	Yearly Server Cost
Hostinger – KVM 2 VPS	2 vCPU, 8 GB RAM, 100 GB NVMe	Rs. 600 – Rs. 800	Rs. 7,200 – Rs. 9,600
DigitalOcean – Basic Droplet	2 vCPU, 4 GB RAM, 80 GB NVMe	\$24 (~Rs. 2,000)	\$288 (~Rs. 24,000)
AWS – Amazon Lightsail	2 vCPU, 4 GB RAM, 80 GB SSD	\$24 (~Rs. 2,000)	\$288 (~Rs. 24,000)

Server Hosting Cost: **Rs. 7,200 – Rs. 24,000** • Domain Cost: **Rs. 499 - Rs. 999 / year** • Total Yearly Cost: **Rs. 7,699 – Rs. 24,999**

Yearly Total Cost Comparison (Server + Domain)

Architecture / Hosting Option	Server Hosting Cost (/yr)	Domain Cost (/yr)	Total Yearly Cost (INR)	Net Result for College
Option 1 On-Campus PC (24×7) College CPU on campus power/internet	Rs. 0 (Free)	Rs. 499 - Rs. 999 / year	Rs. 499 - Rs. 999 / year	Rs. 7,200 – Rs. 24,000+ Saved
Option 2A Hostinger KVM 2 VPS Budget cloud VPS (2 vCPU, 8 GB RAM, 100 GB NVMe)	Rs. 7,200 – Rs. 9,600	Rs. 499 - Rs. 999 / year	Rs. 7,699 – Rs. 10,599	Baseline Budget Reference
Option 2B DigitalOcean / AWS Standard cloud VPS (\$24/mo, 2 vCPU, 4 GB RAM)	Rs. 24,000 (\$288)	Rs. 499 - Rs. 999 / year	Rs. 24,499 – Rs. 24,999	No Savings (Premium SLA)

Strategic Recommendation for BFGI College Leadership

We strongly recommend **Option 1 (On-Campus Dedicated College PC)**. Since the college already maintains campus high-speed internet and 24×7 power backup, deploying one dedicated CPU for the backend incurs **Rs. 0** in recurring server hosting costs. The college only needs to purchase the domain name (Rs. 499 - Rs. 999 / year), resulting in a yearly cost saving of **Rs. 7,200 to Rs. 24,000+** compared to cloud VPS providers, while easily supporting 5,000+ alumni portal users.

■ **Note: Domain registration charges are indicative and may vary from time to time based on registrar pricing, applicable taxes, and renewal policies. Prices should be reconfirmed with the registrar at the time of purchase.**